

OS PRACTICAL

PRACTICAL-3

3. Develop a C program using fork() that creates a parent and child process. Display the Process ID (PID), Parent Process ID (PPID), and process states at different stages of execution.

Design an experiment to observe process state transitions (Ready, Running, Waiting, Terminated) using Linux monitoring tools such as ps, top, and /proc.

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rishitha@Rishitha: ~/practical × + ▾
rishitha@Rishitha:~$ cd ~
rishitha@Rishitha:~$ mkdir practical3
rishitha@Rishitha:~$ cd practical3
rishitha@Rishitha:~/practical3$ pwd
/home/rishitha/practical3
rishitha@Rishitha:~/practical3$ nano practical3.c
rishitha@Rishitha:~/practical3$ gcc practical3.c -o practical3
rishitha@Rishitha:~/practical3$ ./practical3

----- PARENT PROCESS -----
Parent PID : 547

Child PID : 548
----- CHILD PROCESS -----
Child PID : 548
Parent PID : 547
Child Process Ending
Parent Process Ending
rishitha@Rishitha:~/practical3$ cd ~/practical3
rishitha@Rishitha:~/practical3$ ps -ef
UID          PID     PPID  C  STIME TTY          TIME CMD
root           1         0  0   05:21 ?        00:00:01 /sbin/init
root           2         1  0   05:21 hvc0    00:00:00 /init
root           9         2  0   05:21 hvc0    00:00:00 plan9 --control-socket 7
root          52         1  0   05:21 ?        00:00:00 /usr/lib/systemd/systemd
systemd+      69         1  0   05:21 ?        00:00:00 /usr/lib/systemd/systemd
root          94         1  0   05:21 ?        00:00:00 /usr/lib/systemd/systemd
root         148         1  0   05:21 ?        00:00:00 /bin/sh /usr/lib/systemd
root         152         1  0   05:21 ?        00:00:00 /usr/sbin/cron -f -P
message+     155         1  0   05:21 ?        00:00:00 @dbus-daemon --system --
root         166         1  0   05:21 ?        00:00:00 /usr/bin/python3 /usr/bi
root         173         1  0   05:21 ?        00:00:00 /usr/lib/systemd/systemd
root         177         1  0   05:21 ?        00:00:00 /usr/libexec/wsl-pro-ser
syslog       189         1  0   05:21 ?        00:00:00 /usr/sbin/rsyslogd -n -i
_chrony      228        148  0   05:21 ?        00:00:00 /usr/sbin/chronyd -n -F
_chrony      229        228  0   05:21 ?        00:00:00 /usr/sbin/chronyd -n -F
root         276         1  0   05:21 tty1    00:00:00 /usr/sbin/agetty --nores
root         277         1  0   05:21 ?        00:00:00 /usr/bin/python3 /usr/sh
root         338         2  0   05:21 ?        00:00:00 /init
root         339        338  0   05:21 ?        00:00:00 /init
rishitha     340        339  0   05:21 pts/0   00:00:00 -bash
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rishitha@Rishitha: ~/practical x + v
root      341      2  0 05:21 ?        00:00:00 login -- rishitha
rishitha  388      1  0 05:21 ?        00:00:00 /usr/lib/systemd/systemd
rishitha  390     388  0 05:21 ?        00:00:00 (sd-pam)
rishitha  416     341  0 05:21 pts/1    00:00:00 -bash
rishitha  569     340  0 05:23 pts/0    00:00:00 ps -ef
rishitha@Rishitha:~/practical3$ ps -o pid,ppid,state,cmd
  PID   PPID  S  CMD
   340   339  S  -bash
   583   340  R  ps -o pid,ppid,state,cmd
rishitha@Rishitha:~/practical3$ top
top - 05:27:09 up 5 min, 1 user, load average: 0.02, 0.32, 0.21
Tasks: 25 total, 1 running, 24 sleeping, 0 stopped, 0 zombie
%Cpu(s):  0.0 us,  0.0 sy,  0.0 ni,100.0 id,  0.0 wa,  0.0 hi,  0.0 si,  0.
MiB Mem : 7612.6 total, 6456.2 free, 591.9 used, 715.1 buff/cache
MiB Swap: 2048.0 total, 2048.0 free,  0.0 used. 7020.8 avail Mem

  PID USER      PR  NI    VIRT    RES    SHR S  %CPU  %MEM    TIME+
   1 root        20   0    24236   15420   11616 S   0.0   0.2   0:01.24
   2 root        20   0     3180    2208    2072 S   0.0   0.0   0:00.02
   9 root        20   0     3180    2048    1940 S   0.0   0.0   0:00.00
  52 root        19  -1   50400   16544   15272 S   0.0   0.2   0:00.30
  69 systemd+    20   0   22416   14408   11908 S   0.0   0.2   0:00.17
  94 root        20   0   35252   12352    9256 S   0.0   0.2   0:00.61
 148 root        20   0     2888    1944    1820 S   0.0   0.0   0:00.12
 152 root        20   0     4472    3032    2780 S   0.0   0.0   0:00.02
 155 message+    20   0     8692    5288    4624 S   0.0   0.1   0:00.09
 166 root        20   0   44096   29760   14616 S   0.0   0.4   0:00.43
 173 root        20   0   18440    9412    8168 S   0.0   0.1   0:00.23
 177 root        20   0 1866032   15416   12552 S   0.0   0.2   0:00.24
 189 syslog      20   0  220548    5704   4500 S   0.0   0.1   0:00.13
 228 _chrony      20   0   23732   11116    7132 S   0.0   0.1   0:00.08
 229 _chrony      20   0   12096    2424    1780 S   0.0   0.0   0:00.01
 276 root        20   0     5492    2696    2516 S   0.0   0.0   0:00.02
 277 root        20   0  123460   32600   17988 S   0.0   0.4   0:00.30
 338 root        20   0     3184    1112     980 S   0.0   0.0   0:00.00
 339 root        20   0     3200    1252    1108 S   0.0   0.0   0:00.04
 340 rishitha     20   0     6268    5532    3864 S   0.0   0.1   0:00.13
 341 root        20   0     8644    5568    4728 S   0.0   0.1   0:00.01
 388 rishitha     20   0   22360   13424   10792 S   0.0   0.2   0:00.19
 390 rishitha     20   0   22920    4092    2088 S   0.0   0.1   0:00.00
 416 rishitha     20   0     6136    5468    3928 S   0.0   0.1   0:00.04
```

```
rishitha@Rishitha:~/practical3$ cat /proc/548/status
cat: /proc/548/status: No such file or directory
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